

A RATIONAL RANK-(2, 2) HADAMARD FACTORIZATION OF A PROPOSED 4×4 REAL COUNTEREXAMPLE

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ABSTRACT. Rivin identified a binary 4×4 matrix that is not a Hadamard product of two rank-at-most-two matrices over $\mathbb{F}_2$, and reported numerical evidence that the same matrix might remain nonexpressible over $\mathbb{R}$. We give an exact rational factorization of this matrix as $A \circ B$ with $\text{rank } A = \text{rank } B = 2$. More generally, we exhibit a three-parameter family of such factorizations. Thus this particular matrix is not a real counterexample. The result does not decide whether some other invertible real 4×4 matrix fails to admit a rank-(2, 2) Hadamard factorization.

1. INTRODUCTION

For matrices of the same size, their Hadamard product is defined by $(A \circ B)_{ij} = A_{ij}B_{ij}$. The rank inequality

$$\text{rank}(A \circ B) \leq \text{rank}(A) \text{rank}(B)$$

follows immediately by expanding rank factorizations of A and B into rank-one summands. It leaves open the converse-type question of which rank-four matrices are Hadamard products of two matrices of rank at most two.

Rivin [1] studied this question computationally for 4×4 matrices. His Example 5 is

$$(1) \quad M = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

The paper reports an exhaustive search showing that M is not rank-(2, 2) Hadamard expressible over $\mathbb{F}_2$ and reports failed numerical optimization over $\mathbb{R}$ as evidence for real nonexpressibility. Its list of open problems asks for analytic proofs in the real case. The exact calculation below shows that this particular candidate behaves differently over $\mathbb{Q}$ and $\mathbb{R}$ than over $\mathbb{F}_2$.

2. AN EXACT RATIONAL FACTORIZATION

Theorem 2.1. *The matrix M in (1) has a Hadamard factorization $M = A \circ B$ with $A, B \in \mathbb{Q}^{4 \times 4}$ and $\text{rank } A = \text{rank } B = 2$.*

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Proof. Take

$$(2) \quad A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 3 \\ -2 & -1 & 0 & 0 \\ -2 & -1 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \frac{1}{2} & \frac{1}{3} & 0 \\ 0 & -1 & -\frac{4}{3} & -2 \\ -\frac{1}{2} & 0 & \frac{1}{6} & \frac{1}{2} \end{pmatrix}.$$

Entrywise multiplication gives $A \circ B = M$. Writing A_i and B_i for row i , respectively, we have

$$A_3 = A_4 = A_2 - 3A_1, \quad B_3 = 2B_2 - 2B_1, \quad B_4 = -B_2 + \frac{1}{2}B_1.$$

Hence both ranks are at most two. The minors in rows 1, 2 and columns 1, 2 are 1 for A and $-\frac{1}{2}$ for B , so both ranks are exactly two. Finally, direct expansion gives $\det M = 1$, confirming that the target matrix is invertible. $\square$

This certificate uses only rational arithmetic. In particular, it is not a rational reconstruction of a merely approximate equality.

3. A THREE-PARAMETER FAMILY

The factorization in Theorem 2.1 arises from equality fibres that distribute the zero pattern of M between the two factors. The following family makes this structure explicit.

Proposition 3.1. *Let x, y, z be pairwise distinct nonzero real numbers. Put*

$$\mathbf{1} = (1, 1, 1, 1), \quad a = (x, y, z, z), \quad b = (x^{-1}, y^{-1}, z^{-1}, 0),$$

and define the rows of A and B by

$$\begin{aligned} A_1 &= B_1 = \mathbf{1}, & A_2 &= a, & B_2 &= b, \\ A_3 &= A_4 = a - z\mathbf{1}, & B_3 &= \frac{xy}{(y-z)(x-y)}(b - x^{-1}\mathbf{1}), \\ & & B_4 &= \frac{xy}{(x-z)(y-x)}(b - y^{-1}\mathbf{1}). \end{aligned}$$

Then $A \circ B = M$ and $\text{rank } A = \text{rank } B = 2$. If $x, y, z \in \mathbb{Q}$, the factors are rational.

Proof. All displayed denominators are nonzero by hypothesis. Every row of A lies in $\text{span}\{\mathbf{1}, a\}$, and every row of B lies in $\text{span}\{\mathbf{1}, b\}$, so both ranks are at most two. Direct entrywise multiplication gives

$$\begin{aligned} A_1 \circ B_1 &= (1, 1, 1, 1), & A_2 \circ B_2 &= (1, 1, 1, 0), \\ A_3 \circ B_3 &= (0, 1, 0, 0), & A_4 \circ B_4 &= (1, 0, 0, 0). \end{aligned}$$

For instance, the only nonzero target entry in the third row follows from

$$(y-z) \frac{xy}{(y-z)(x-y)} (y^{-1} - x^{-1}) = 1;$$

the fourth row is analogous. Pairwise distinctness implies that $\mathbf{1}, a$ are independent, while the last coordinate of b is zero and its other coordinates are nonzero, so $\mathbf{1}, b$ are independent. Thus both ranks equal two. $\square$

The specialization $(x, y, z) = (1, 2, 3)$ gives (2). The family also identifies a branch that a zero-pattern infeasibility search must retain: both third and fourth rows of A vanish in columns 3, 4, while the corresponding rows of B vanish in columns 1 and 2, respectively.

4. SCOPE AND REPRODUCIBILITY

Theorem 2.1 refutes only the assertion that the specific matrix (1) is nonexpressible over $\mathbb{R}$. It neither proves nor disproves the universal statement that every invertible real 4×4 matrix admits a rank-(2,2) Hadamard factorization. Nor does a rational witness contradict an integer-only nonexpressibility statement: the factors in (2) are not both integral.

The project archive contains the serialized exact matrices, a standard-library Python verifier, and stable reproduction commands. The verifier reconstructs every entrywise product, checks all 3×3 minors of both factors, exhibits nonzero 2×2 minors, and checks $\det M$. A second independently derived certificate and verifier are also included. No floating-point optimization or random seed is involved in the proof.

COMPUTATIONAL AND AI DISCLOSURE

The factorization search, symbolic checks, verifier generation, literature search assistance, and initial drafting used OpenAI Codex with multiple adversarial agent roles. The mathematical endpoint is the exact rational identity displayed in (2); it can be checked without trusting the discovery computation or an AI system. No proof assistant was used.

REFERENCES

1. Igor Rivin, *Computational resolution of hadamard product factorization for 4×4 matrices*, 2025, arXiv:2508.14901, version 1, submitted July 31, 2025.

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